In [2]:

```python
# Data cleaning process

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the datasets
train_df = pd.read_csv(r'C:\Users\LOQ\Downloads\train.csv')
test_df = pd.read_csv(r'C:\Users\LOQ\Downloads\test.csv')

# Display initial missing value counts
print("Missing values in Training Data before cleaning:")
print(train_df.isnull().sum())
print("-" * 40)

# 2. Handle missing values for 'Age' (Fill with Median)
train_df['Age'] = train_df['Age'].fillna(train_df['Age'].median())
test_df['Age'] = test_df['Age'].fillna(test_df['Age'].median())

# 3. Handle missing values for 'Embarked' (Fill with Mode/Most Frequent)
most_frequent_embarked = train_df['Embarked'].mode()[0]
train_df['Embarked'] = train_df['Embarked'].fillna(most_frequent_embarked)

# 4. Handle missing values for 'Fare' in the test dataset (Fill with Median)
test_df['Fare'] = test_df['Fare'].fillna(test_df['Fare'].median())

# 5. Drop the 'Cabin' column (Too many missing values to accurately fill)
train_df.drop(columns=['Cabin'], inplace=True, errors='ignore')
test_df.drop(columns=['Cabin'], inplace=True, errors='ignore')

print("Data Cleaning Complete. Remaining missing values in Train:", train_df.isnull
```

```
Missing values in Training Data before cleaning:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
----------------------------------------
Data Cleaning Complete. Remaining missing values in Train: 0
```

In [3]:

```python
# Data Analysis (EDA)

# Group by Gender to calculate survival probability
gender_survival = train_df[['Sex', 'Survived']].groupby(['Sex'], as_index=False).me
print("\n--- Survival Rate by Gender ---")
print(gender_survival)
```

```
# Group by Passenger Class to calculate survival probability
class_survival = train_df[['Pclass', 'Survived']].groupby(['Pclass'], as_index=Fals
print("\n--- Survival Rate by Passenger Class ---")
print(class_survival)

# Group by Family Size (SibSp + Parch) to see if traveling together helped
train_df['FamilySize'] = train_df['SibSp'] + train_df['Parch'] + 1
family_survival = train_df[['FamilySize', 'Survived']].groupby(['FamilySize'], as_i
print("\n--- Survival Rate by Family Size ---")
print(family_survival)
```

```
--- Survival Rate by Gender ---
      Sex  Survived
0  female  0.742038
1    male  0.188908

--- Survival Rate by Passenger Class ---
   Pclass  Survived
0       1  0.629630
1       2  0.472826
2       3  0.242363

--- Survival Rate by Family Size ---
   FamilySize  Survived
0           1  0.303538
1           2  0.552795
2           3  0.578431
3           4  0.724138
4           5  0.200000
5           6  0.136364
6           7  0.333333
7           8  0.000000
8          11  0.000000
```

In [6]:

```
# Data visualization

# Set up the visualization style
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Plot 1: Survival Rate by Gender
sns.barplot(ax=axes[0], x='Sex', y='Survived', data=train_df, errorbar=None)
axes[0].set_title('Survival Rate by Gender')
axes[0].set_ylabel('Proportion of Survivors')

# Plot 2: Survival Rate by Passenger Class
sns.barplot(ax=axes[1], x='Pclass', y='Survived', data=train_df, errorbar=None)
axes[1].set_title('Survival Rate by Passenger Class')
axes[1].set_ylabel('Proportion of Survivors')

# Plot 3: Age Distribution of Survivors vs Non-Survivors
sns.histplot(ax=axes[2], data=train_df, x='Age', hue='Survived', multiple='stack',
axes[2].set_title('Age Distribution by Survival Status')
axes[2].set_xlabel('Age')
```

```
plt.tight_layout()
plt.show()
```